DevDocs AI - Database and Vector Index Architecture Spec

Section 1: MongoDB Atlas Multi-Region Cluster

The DevDocs AI storage engine is hosted on MongoDB Atlas M10 dedicated cluster across three availability zones.
Documentation metadata is organized into collection 'documents' and text chunks are stored in 'chunks'.
Each document record tracks document_id, document_name, upload_date, owner, and content_type.
Each chunk record contains chunk_id, document_id, page_number, text, and vector embedding.
Vector search is executed natively in MongoDB Atlas using the specialized $vectorSearch aggregation operator.
The vector index is designated as 'chunks_vector_index' built on the 'embedding' field.
The embedding model is sentence-transformers/all-MiniLM-L6-v2 outputting exactly 384 dimensions.
The index metric type is cosine similarity with 150 candidates evaluated per vector search query.

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Section 2: Hybrid Search and Benchmark Evaluation

To capture both broad semantic intent and exact code symbols, DevDocs AI implements Hybrid Search.
Semantic vector retrieval is fused with lexical BM25/regex matching using Reciprocal Rank Fusion (RRF).
The RRF constant k is calibrated to 60, allowing balanced combination of dense vectors and exact keywords.
Retrieval performance is continuously measured against ground-truth technical queries.
Evaluation metrics include Hit Rate@5, Precision@5, and Recall@5.
Target benchmark standards require a Hit Rate@5 greater than 0.85 across benchmark queries.
Cloud snapshot backups are automatically executed every 24 hours with a rolling 7-day retention policy.